

WEIDE LIU

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Profile Summary

Weide Liu, Full Professor at the School of Computing and Artificial Intelligence, **Jiangxi University of Finance and Economics (JUFE)**. He received both his B.Eng. in Electrical and Electronic Engineering and his Ph.D. in Computer Science from **Nanyang Technological University (NTU)**, Singapore.

Prior to joining JUFE, he was a **Research Fellow** at **Harvard Medical School** and at **Nanyang Technological University (NTU)**, and a Research Scientist at the Agency for Science, Technology and Research (A*STAR) and at ByteDance AI Lab, Singapore.

His research spans artificial intelligence, computer vision, medical image analysis, imaging neuroscience, time-series analysis, multimodal learning and machine learning methodology. He has published **60+** papers in leading international conferences and journals including CVPR, ICCV, ECCV, ICML, IJCV and IEEE TIP/TMI/TMM, and has received the **Best Paper Award** at ICAIS&ISAS 2026, the **Best Paper Award** at the “International Workshop on Trustworthy Multimodal Learning for Social Media Analysis,” The ACM Web Conference 2026, and the **Best Paper Award** at the “Deep Learning for Human Activity Recognition” workshop, IJCAI 2024.

Professional service: Associate Editor of *IEEE Journal of Biomedical and Health Informatics (JBHI)* and *Signal, Image and Video Processing*; Special Session Chair for IEEE ICIP 2024–2026 and IEEE ICME 2025–2026; Area Chair for top-tier international conferences including BMVC and AAAI. Recipient of the **DAAD AInet Fellowship 2024** (AI for Science).

Research Interests

Centred on **AI for healthcare**, **computer vision** and **time-series analysis**, combined with machine learning methodology, addressing both fundamental scientific questions and clinical applications:

- **Fundamental computer vision:** image segmentation, restoration and enhancement, representation learning, few-shot and weakly supervised learning, long-tailed recognition.
- **Medical imaging and imaging neuroscience:** fetal brain development imaging (diffusion MRI, tractography, cross-modality registration), intelligent disease analysis, medical image quality assessment.
- **Computational pathology:** multi-scale representation and multiple-instance weakly supervised learning for whole-slide images, quantitative analysis of nuclei and the tissue microenvironment, stain normalisation and cross-centre domain generalisation, pathology–genomics–clinical multimodal fusion for subtyping and prognosis, pathology foundation models and structured report generation.
- **Trustworthy AI and causal inference:** evidential deep learning, Bayesian uncertainty modelling, causal discovery and causal feature learning.
- **Multimodal learning and time-series analysis:** cross-modal alignment, vision–language models, time-series modelling, federated learning and unsupervised domain adaptation.

Education

Nanyang Technological University (NTU), Singapore
Ph.D., School of Computer Science and Engineering

Aug. 2018 – Jun. 2022

Nanyang Technological University (NTU), Singapore
B.Eng., Electrical and Electronic Engineering. Full scholarship (Nanyang Scholarship).

Jul. 2012 – Jun. 2016

Work Experience

Jiangxi University of Finance and Economics (JUFE) | Full Professor Dec. 2025 – Present

- Conducting frontier research in imaging neuroscience, medical image analysis and AI for Science.
- Promoting interdisciplinary collaboration; supervising graduate students and junior researchers in computer vision, biomedical engineering and time-series intelligence.
- Supporting AI curriculum development, research platform construction and talent programme cultivation, serving the university's strategic development in artificial intelligence and digital innovation.

Harvard Medical School | Research Fellow

Jan. 2024 – Dec. 2025

- Developed motion-robust fetal brain imaging analysis techniques for imaging neuroscience, including machine-learning-based tractography and cross-modality registration methods, applied to brain development assessment in utero and in early life.
- Collaborated closely with the Computational Radiology Laboratory (CRL) team at Harvard Medical School (S. Warfield, D. Karimi, P. E. Grant, et al.), translating methodological advances into deployable tools for early brain development research and clinical decision support.

Agency for Science, Technology and Research (A*STAR), Singapore | Research Scientist
Aug. 2021 – Jan. 2024

- Research covering video segmentation, weakly supervised segmentation, 3D point cloud analysis, few-shot learning, long-tailed recognition and multimodal recognition.
- Led and participated in multiple industry-academia collaboration projects, publishing in top-tier computer vision and multimedia venues (CVPR, ICCV, IJCV, IEEE TMM, etc.).

ByteDance AI Lab | Research Scientist

Dec. 2020 – Aug. 2021

- Worked on multimedia content understanding and computer vision, contributing to algorithm development for industrial-scale AI systems.

Selected Publications

60+ peer-reviewed papers (CVPR / ICCV / ECCV / ICML / AAAI / IJCAI / IJCV / IEEE TIP / TMI / TMM, etc.), grouped thematically below.

* Corresponding Author; † Equal Contribution

Impact factors (IF) are based on JCR 2025.

I. Efficient Learning

1. **Weide Liu**, Chi Zhang, Guosheng Lin, Fayao Liu, "CRCNet: Few-shot Segmentation with Cross-Reference and Region-Global Conditional Networks," *International Journal of Computer Vision*, vol. 130, no. 12, pp. 3140–3157, 2022. DOI: 10.1007/s11263-022-01677-7. (Top Journal, IF: 9.3)

2. **Weide Liu**, Zhonghua Wu, Yang Zhao, Yuming Fang, Chuan-Sheng Foo, Jun Cheng, Guosheng Lin, "Harmonizing Base and Novel Classes: A Class-Contrastive Approach for Generalized Few-Shot Segmentation," *International Journal of Computer Vision*, vol. 132, no. 4, pp. 1277–1291, April 2024. DOI: 10.1007/s11263-023-01939-y. (Top Journal, IF: 9.3)
3. **Weide Liu**, Xiaoyang Zhong, Lu Wang, Chunbo Lang, Yuming Fang, Jun Cheng, Xulei Yang and Gong Cheng, "Evidential Robust Feature Learning for Generalized Few-Shot Segmentation," *International Journal of Computer Vision*, vol. 134, article number 209, 2026. DOI: 10.1007/s11263-026-02796-1. (Top Journal, IF: 9.3)
4. **Weide Liu**, Chi Zhang, Guosheng Lin, Fayao Liu, "CRNet: Cross-Reference Networks for Few-Shot Segmentation," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020, pp. 4164–4172. (Acceptance rate: 22%)
5. **Weide Liu**, Xiangfei Kong, Tzu-Yi Hung, Guosheng Lin, "Cross-Image Region Mining with Region Prototypical Network for Weakly Supervised Segmentation," *IEEE Transactions on Multimedia*, vol. 25, pp. 1148–1160, 2023. DOI: 10.1109/TMM.2021.3139459. (Top Journal, IF: 9.7)
6. **Weide Liu**, Chi Zhang, Henghui Ding, Tzu-Yi Hung, Guosheng Lin, "Few-Shot Segmentation with Optimal Transport Matching and Message Flow," *IEEE Transactions on Multimedia*, vol. 25, pp. 5130–5141, 2023. DOI: 10.1109/TMM.2022.3187855. (Top Journal, IF: 9.7)
7. **Weide Liu**[†], Zhonghua Wu[†], Yiming Wang, Henghui Ding, Fayao Liu, Jie Lin, Guosheng Lin, "LCReg: Long-Tailed Image Classification with Latent Categories based Recognition," *Pattern Recognition*, vol. 145, article 109971, February 2024. DOI: 10.1016/j.patcog.2023.109971. (Top Journal, IF: 7.6)
8. **Weide Liu**, Chi Zhang, Guosheng Lin, Tzu-Yi Hung, Chunyan Miao, "Weakly Supervised Segmentation with Maximum Bipartite Graph Matching," in *Proceedings of the 28th ACM International Conference on Multimedia (ACM MM)*, 2020, pp. 2085–2094. DOI:10.1145/3394171.3413652 (Acceptance rate: 27%)
9. Tianyi Zhang, Guosheng Lin, **Weide Liu**, Jianfei Cai, Alex Kot, "Splitting vs. Merging: Mining Object Regions with Discrepancy and Intersection Loss for Weakly Supervised Semantic Segmentation," in *Proceedings of the European Conference on Computer Vision (ECCV)*, 2020, pp. 663–679. (Acceptance rate: 27%)
10. **Weide Liu**, Jieming Lou, Xingxing Wang, Wei Zhou, Jun Cheng, Xulei Yang, "Physically-guided Open Vocabulary Segmentation with Weighted Patched Alignment Loss," *Neurocomputing*, vol. 614, article 128788, January 2025. DOI: 10.1016/j.neucom.2024.128788. (Top Journal, IF: 6.5)
11. **Weide Liu**, Zhonghua Wu, Henghui Ding, Fayao Liu, Jie Lin, Guosheng Lin, Wei Zhou, "Exploiting independent query information for few-shot image segmentation," *Displays*, vol. 91, article 103179, 2026.
12. Lechun You, Zhonghua Wu, **Weide Liu**^{*}, Xulei Yang, Jun Cheng, Wei Zhou, Bharadwaj Veeravalli, Guosheng Lin, "Integrating SAM Supervision for 3D Weakly Supervised Point Cloud Segmentation," *IEEE Transactions on Image Processing (TIP)*, 2026. DOI: 10.1109/TIP.2026.3691686. (Top Journal, IF: 13.7)
13. Guanghui Yue, Xiangfei Meng, Beibei Li, Tianwei Zhou, Danling Jiang, Juerong Feng, **Weide Liu**^{*}, Yongfei Wu, Haoren Xie, Jun Cheng, "Enhancing Pseudo-Label Reliability for Semi-Supervised Gastric Polyp Segmentation," *IEEE Transactions on Image Processing (TIP)*, 2026. DOI: 10.1109/TIP.2026.3731602. (Top Journal, IF: 13.7)

II. Medical Imaging & AI for Neuroscience

14. **Weide Liu**, Camilo Calixto, Simon K. Warfield, Davood Karimi, “Streamline tractography of the fetal brain in utero with machine learning,” *Imaging Neuroscience*, vol. 3, 2025. DOI: 10.1162/imag_a_00537. (Top Journal)
15. Qi Zeng, **Weide Liu**, Bo Li, Ryne Didier, P. Ellen Grant, Davood Karimi, “Towards automatic US-MR fetal brain image registration with learning-based methods,” *NeuroImage*, vol. 310, article 121104, doi: 10.1016/j.neuroimage.2025.121104, 2025. (Top Journal)
16. **Weide Liu**, Jingwen Hou, Xiaoyang Zhong, Huijing Zhan, Jun Cheng, Yuming Fang, Guanghui Yue, “Improving Multi-Modal Brain Tumor Segmentation via Pre-Training and Knowledge Distillation Based Post-Training,” *Neurocomputing*, vol. 640, article 130318, August 2025. DOI: 10.1016/j.neucom.2025.130318. (Top Journal, IF: 6.5)
17. Guanghui Yue, Shaoping Zhang, Tianwei Zhou, Bin Jiang, **Weide Liu***, Tianfu Wang, “Pyramid Network with Quality-Aware Contrastive Loss for Retinal Image Quality Assessment,” *IEEE Transactions on Medical Imaging (TMI)*, vol. 44, no. 3, pp. 1416–1431, 2025, doi: 10.1109/TMI.2024.3501405. (Top Journal, IF: 9.8)
18. Guanghui Yue, Di Cheng, Tianwei Zhou, Jingwen Hou, **Weide Liu**, Long Xu, Tianfu Wang, Jun Cheng, “Perceptual Quality Assessment of Enhanced Colonoscopy Images: A Benchmark Dataset and an Objective Method,” *IEEE Transactions on Circuits and Systems for Video Technology*, vol. 33, no. 10, pp. 5549–5561, October 2023, doi: 10.1109/TCSVT.2023.3260212. (Top Journal, IF: 11.1)
19. Guanghui Yue, Guibin Zhuo, Siying Li, Tianwei Zhou, Jingfeng Du, Weiqing Yan, Jingwen Hou, **Weide Liu**, Tianfu Wang, “Benchmarking Polyp Segmentation Methods in Narrow-Band Imaging Colonoscopy Images,” *IEEE Journal of Biomedical and Health Informatics (JBHI)*, vol. 27, no. 7, pp. 3360–3371, July 2023, doi: 10.1109/JBHI.2023.3270724. (Top Journal, IF: 6.8)
20. Guanghui Yue, Guibin Zhuo, Tianwei Zhou, **Weide Liu**, Tianfu Wang, Qiuping Jiang, “Adaptive Cross-Feature Fusion Network With Inconsistency Guidance for Multi-Modal Brain Tumor Segmentation,” *IEEE Journal of Biomedical and Health Informatics (JBHI)*, vol. 29, no. 5, pp. 3148–3158, May 2025, doi: 10.1109/JBHI.2023.3347556. (Top Journal, IF: 6.8)
21. Guanghui Yue, Peishan Wei, Tianwei Zhou, Chang Tang, **Weide Liu**, Jie Du, “FedPDN: Personalized Federated Learning with Inter-class Similarity Constraint for Medical Image Classification through Parameter Decoupling,” *IEEE Transactions on Instrumentation and Measurement (IEEE TIM)*, 2025. (Top Journal, IF: 5.9)
22. Zhantao Cao, Yuanbing Shi, Shuli Zhang, Huanan Chen, **Weide Liu***, Guanghui Yue, Huazhen Lin, “Decentralized learning for medical image classification with prototypical contrastive network,” *Medical Physics*, vol. 52, no. 6, pp. 4188–4204, June 2025, DOI: 10.1002/mp.17753. (Top Journal, IF: 3.8)
23. Tianrui Liu, Qiyue Wei, Jianguo Chen, **Weide Liu**, Weimin Huang, Ruchir Srivastava, Zhongyao Cheng, Zeng Zeng, Bharadwaj Veeravalli, Xulei Yang, “Unsupervised 3D Lung Segmentation by Leveraging 2D Segment Anything Model,” in *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2024, pp. 1–4.
24. Shenghao Zhao, Xiaoyu Zhang, Ziyuan Zhao, Peisheng Qian, **Weide Liu**, Zeng Zeng, Bharadwaj Veeravalli, Lingyun Dai, Pär Nordlund, Nayana Prabhu, Wai Leong Tam, Xulei Yang, “Hybrid Model Design For Protein Function Prediction,” in *Proceedings of the Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, 2024, pp. 1–4.
25. Honghui Chen, Baoquan Zhao, Guanghui Yue, **Weide Liu**, Chenlei Lv, Ruomei Wang, Fan Zhou, “CLIP-MEDFAKE: Synthetic Data Augmentation with AI-Generated Content for Improved Medical Image Classification,” in *Proceedings of the IEEE International Conference on Image Processing*

(ICIP), 2024, pp. 3854–3860.

26. Jian Wang, Sixing Rong, Jiarui Xing, Yuling Xu and **Weide Liu***, “PathoSyn: Imaging-Pathology MRI Synthesis via Disentangled Deviation Diffusion”, *IEEE Journal of Biomedical and Health Informatics (JBHI)*, 2026. (Top Journal, IF: 6.8)
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III. Causal Inference & Machine Learning Theory

27. Liang Cao, Jun Wan, Yan Qin, **Weide Liu***, “Causal Feature Learning via Generalized Rayleigh Quotients,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2026.
 28. Liang Cao, **Weide Liu***, Zhuo Chen, Yan Qin, “A Control-Theoretic View of Mamba on Stability and Robustness,” in *Proceedings of the International Conference on Machine Learning (ICML)*, 2026.
 29. Liang Cao, **Weide Liu***, Yang Liu, Yan Qin, “Condition-Dependent Causal Discovery: A Polynomial Chaos Framework for Systems with Parametric Uncertainty,” *IEEE Transactions on Automation Science and Engineering (T-ASE)*, vol. 23, pp. 6287–6298, 2026, doi: 10.1109/TASE.2026.3672332. (Top Journal, IF: 6.8)
 30. Liang Cao, **Weide Liu***, Yan Qin, “Causal Discovery in Dynamic Industrial Systems Under Parametric Uncertainty: A Polynomial Chaos Approach,” *IEEE Transactions on Industrial Informatics*, 2026, doi: 10.1109/TII.2026.3672049. (Top Journal, IF: 11.7)
 31. Liang Cao, **Weide Liu***, Yan Qin, Zhenghua Chen, Zhuo Chen, Bhushan Gopaluni, “Sparse Causal Latent Features for Robust Multimodal Learning under Distribution Shifts”, 2026, WWW Companion '26: Companion Proceedings of the ACM Web Conference 2026, doi: 10.1145/3774905.3795844.
 32. Liang Cao, **Weide Liu***, Yunhong Che, Yankai Cao, Fan Yang, Bhushan Gopaluni, “Nonlinear Causal Sparse-Smooth Learning for Interpretable Industrial Process Modeling”, 2026, ICAIS&ISAS, (**Best Paper Award**)
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IV. Multimodal Learning & Stereo Vision

33. **Weide Liu**, Xiaoyang Zhong, Lu Wang, Jingwen Hou, Yuemei Luo, Jiebin Yan, Yuming Fang, “Uncertainty Awareness for Unsupervised Domain Adaptation on Human Activity Recognition,” *International Joint Conference on Artificial Intelligence (IJCAI)*, 2024.
34. **Weide Liu**, Xiaoyang Zhong, Lu Wang, Jingwen Hou, Yuemei Luo, Jiebin Yan, and Yuming Fang, “Uncertainty Awareness on Unsupervised Domain Adaptation for Time Series Data,” *IEEE Transactions on Multimedia (TMM)*, DOI: 10.1109/TMM.2025.3639929, 2025. (Top Journal, IF: 9.7)
35. **Weide Liu**, Xiaoyang Zhong, Jingwen Hou, Shaohua Li, Haozhe Huang, Wei Zhou, Yuming Fang, “Integrating Large Foundation Models into Multimodal Named Entity Recognition with Evidential Fusion,” *Neurocomputing*, vol. 652, article 131015, November 2025. DOI: 10.1016/j.neucom.2025.131015. (Top Journal, IF: 6.5)
36. **Weide Liu**, Huijing Zhan, “Multimodal Sentiment Analysis with Missing Modality: A Knowledge-Transfer Approach,” *Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, 2025, pp. 1476–1480.
37. Rui Gong, **Weide Liu**, Zaiwang Gu, Xulei Yang, Jun Cheng, “Learning Intra-view and Cross-view Geometric Knowledge for Stereo Matching,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024, pp. 20752–20762, doi:10.1109/CVPR52733.2024.01961.
38. Jieming Lou, **Weide Liu**, Zhuo Chen, Fayao Liu, Jun Cheng, “ELFNet: Evidential Local-global

- Fusion for Stereo Matching,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023, pp. 17738–17747.
39. Rui Gong, Kim-Hui Yap, **Weide Liu**, Xulei Yang, Jun Cheng, “Rectification-specific Supervision and Constrained Estimator for Online Stereo Rectification,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 22348–22358.
 40. Jianyang Zhang, Qianli Luo, Guowu Yang, Wenjing Yang, **Weide Liu**, Guosheng Lin, Fengmao Lv, “Attribute-formed Class-specific Concept Space: Endowing Language Bottleneck Model with Better Interpretability and Scalability,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025, pp. 30291–30300.
 41. Jiayuan Fan, Haixiang Chen, **Weide Liu**, Xun Xu, Jun Cheng, “Holistic and Contextual Evidential Stereo-LiDAR Fusion for Depth Estimation,” *IEEE Transactions on Intelligent Vehicles (TIV)*, vol. 9, no. 11, pp. 7437–7448, 2024, doi: 10.1109/TIV.2024.3398210. (Top Journal, IF: 14)
 42. Yu Wang, Xiaoye Wang, Zaiwang Gu, **Weide Liu**, Wee Siong Ng, Weimin Huang, and Jun Cheng, “SuperJunction: Learning-Based Junction Detection for Retinal Image Registration,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2024, pp. 292–300. (Top Conference)
 43. Huijing Zhan, Ling Li, Shaohua Li, **Weide Liu**, Manas Gupta, Alex C. Kot, “Towards Explainable Recommendation via BERT-guided Explanation Generator,” in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023, pp. 1–5, doi: 10.1109/ICASSP49357.2023.10094765. (Top Conference)
 44. Jingbo Zeng, Zaiwang Gu, **Weide Liu**, Lile Cai, and Jun Cheng, “Uncertainty Aware Interest Point Detection and Description,” in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2025, pp. 2144–2153. (Top Conference)
 45. Yang Yu, Jiahao Wang, **Weide Liu**, Ivan Ho Mien, Pavitra Krishnaswamy, Xulei Yang, Jun Cheng, “Multimodal multitask similarity learning for vision language model on radiological images and reports,” *Neurocomputing*, vol. 636, article 130018, July 2025, doi: 10.1016/j.neucom.2025.130018. (Top Journal, IF: 6.5)
 46. Hongjun Li, Nan Wang, Anqing Chen, Jiang Liu, Wanli Ma, **Weide Liu**, Yakun Ju, Paul L. Rosin, Hantao Liu, Wei Zhou, “EHIN: Early-aware Hierarchical Interaction Network for Weakly-Supervised Referring Image Segmentation,” *Neurocomputing*, vol. 659, article 131764, 2026. DOI: 10.1016/j.neucom.2025.131764. (Top Journal, IF: 6.5)
 47. Lingli Zhang, Tianrui Li, Baiyu Lu, Junlin Fang, Desheng Zheng, Wei Zhou, **Weide Liu**, Fengmao Lv, “Rethinking the Effect of Unimodal Labels in Multimodal Sentiment Analysis,” *ACM Transactions on Multimedia Computing, Communications and Applications (ACM TOMM)*, 2026. (Top Journal, accepted)
 48. Boan Chen, Liang Cao, **Weide Liu**, Jinmei Liu, Baoquan Zhao, Yang Liu, “Cross-modal attention fusion of RGB and skeleton for multimodal-driven video anomaly detection,” *Pattern Recognition*, vol. 179, article 113815, 2026. DOI: 10.1016/j.patcog.2026.113815. (Top Journal, IF: 7.6)

V. Image Quality Assessment, Video Analysis & Federated Learning

49. **Weide Liu**, Guosheng Lin, Tianyi Zhang, and Zichuan Liu, “Guided Co-Segmentation Network for Fast Video Object Segmentation,” *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, vol. 31, no. 4, pp. 1607–1617, 2021, doi: 10.1109/TCSVT.2020.3010293. (Top Journal, IF: 11.1)
50. Chao Cai, **Weide Liu***, Xue Xia, Zhenghua Chen, and Yuming Fang, “Bayesian Uncertainty Calibration for Federated Time Series Analysis,” *IEEE Transactions on Multimedia*, vol. 26, pp.

- 11151–11163, 2024, doi: 10.1109/TMM.2024.3443627. (Top Journal, IF: 9.7)
51. Chao Cai, Yuming Fang, **Weide Liu***, Ruibing Jin, Jun Cheng, and Zhenghua Chen, “Fed-Cov: Enhanced Trustworthy Federated Learning for Machine RUL Prediction with Continuous-to-Discrete Conversion,” *IEEE Transactions on Industrial Informatics*, vol. 20, no. 11, pp. 12565–12574, Nov. 2024, doi: 10.1109/TII.2024.3423314. (Top Journal, IF: 11.7)
 52. Lisha Li, Jingwen Hou, **Weide Liu**, Yuming Fang, Jiebin Yan, “Diffusion-Based Facial Aesthetics Enhancement with 3D Structure Guidance,” *IEEE Transactions on Image Processing (TIP)*, vol. 34, pp. 1879–1894, 2025, doi: 10.1109/TIP.2025.3551077. (Top Journal, IF: 13.7)
 53. Jingwen Hou, Weisi Lin, Guanghui Yue, **Weide Liu**, Baoquan Zhao, “Interaction-Matrix Based Personalized Image Aesthetics Assessment,” *IEEE Transactions on Multimedia (TMM)*, vol. 25, pp. 5263–5278, 2023, doi: 10.1109/TMM.2022.3189276. (Top Journal, IF: 9.7)
 54. Jingwen Hou, Henghui Ding, Weisi Lin, **Weide Liu**, Yuming Fang, “Distilling Knowledge from Object Classification to Aesthetics Assessment,” *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, vol. 32, no. 11, pp. 7386–7402, 2022, doi: 10.1109/TCSVT.2022.3186307. (Top Journal, IF: 11.1)
 55. Jingwen Hou, Weisi Lin, Yuming Fang, Haoning Wu, Chaofeng Chen, Liang Liao, **Weide Liu**, “Toward Transparent Deep Image Aesthetics Assessment With Tag-Based Content Descriptors,” *IEEE Transactions on Image Processing (TIP)*, vol. 34, pp. 3070–3085, 2025, doi: 10.1109/TIP.2023.3308852. (Top Journal, IF: 13.7)
 56. Jiebin Yan, Lei Wu, Yuming Fang, Xuelin Liu, Xue Xia, and **Weide Liu**, “Video Quality Assessment for Online Processing: From Spatial to Temporal Sampling,” *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, vol. 34, no. 12, pp. 13441–13451, December 2024, DOI: 10.1109/TCSVT.2024.3450085. (Top Journal, IF: 11.1)
 57. Zaiwang Gu, **Weide Liu**, Xulei Yang, Chuan-Sheng Foo, and Jun Cheng, “When Low-Vision Task Meets Dense Prediction Tasks with Less Data: An Auxiliary Self-Trained Geometry Regularization,” *Transactions on Machine Learning Research (TMLR)*, 2024. (Top Journal)
 58. Jiebin Yan, Jiale Rao, Junjie Chen, Ziwen Tan, **Weide Liu**, and Yuming Fang, “Multitask Auxiliary Network for Perceptual Quality Assessment of Non-Uniformly Distorted Omnidirectional Images,” *IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)*, vol. 35, no. 3, pp. 2782–2793, 2025. (Top Journal, IF: 11.1)
 59. Jiebin Yan, Jiale Rao, Xuelin Liu, Yuming Fang, Yifan Zuo, and **Weide Liu**, “Subjective and Objective Quality Assessment of Non-Uniformly Distorted Omnidirectional Images,” *IEEE Transactions on Multimedia (TMM)*, vol. 27, pp. 2695–2707, 2025. (Top Journal, IF: 9.7)
 60. Jiebin Yan, Kangcheng Wu, Junjie Chen, Ziwen Tan, Yuming Fang, **Weide Liu**, “Viewport-unaware Blind Omnidirectional Image Quality Assessment: A Flexible and Effective Paradigm,” *ACM Transactions on Multimedia Computing, Communications and Applications (ACM TOMM)*, vol. 21, no. 5, article 145, pp. 145:1–145:19, 2025. DOI: 10.1145/3723165. (Top Journal, IF: 6.0)
 61. Yixiao Li, Xiaoyuan Yang, **Weide Liu**, Xin Jin, Xu Jia, Yukun Lai, Haotao Liu, Paul L Rosin, Wei Zhou, “TINQ: Temporal inconsistency guided blind video quality assessment,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2026. (Top Conference)
 62. Jun Cheng, Zaiwang Gu, **Weide Liu**, Jiayuan Fan, Zhengguo Li, Chuan-Sheng Foo, “SRNet: Self-supervised Structure Regularization for Stereo Matching,” *Neurocomputing*, vol. 661, article 131907, January 2026. DOI: 10.1016/j.neucom.2025.131907. (Top Journal, IF: 6.5)
 63. Yufeng Chen, Guanghui Yue, **Weide Liu**, Chenlei Lv, Ruomei Wang, Fan Zhou, Baoquan Zhao, “Predicting Plain Text Imageability for Faithful Prompt-Conditional Image Generation,” in *PRI-*

CAI 2024: Trends in Artificial Intelligence, Lecture Notes in Computer Science, vol. 15283, pp. 89–95, Springer, 2025. DOI: 10.1007/978-981-96-0122-6_9.

64. Jingwen Hou, Zengliang Li, Jiebin Yan, Weide Liu, Yuming Fang, Wei Zhou, “Frequency-Aware Native Resolution Assessment of 8K Omnidirectional Images”, in *2025 International Conference on Visual Communications and Image Processing (VCIP)*, pp. 1–5. DOI:10.1109/VCIP67698.2025.11396856.
65. Yixiao Li, ..., Weide Liu, “VQualA 2025 Challenge on Image Super-Resolution Generated Content Quality Assessment: Methods and Results”, in *IEEE/CVF International Conference on Computer Vision Workshops (ICCVW)*, 2025, pp. 3372–3382.

VI. Others

66. **Weide Liu**, Wei Zhou, Jun Liu, Ping Hu, Jun Cheng, Jungong Han, Weisi Lin, “Modality-Aware Feature Matching in Visual and Vision-Language Applications: A Comprehensive Survey”, *ACM Computing Surveys*, 2026. (Top Journal, **IF: 28.0**)
67. Tao Li, Liang Cao, Kai Zhao, **Weide Liu**, Yan Qin, “Capacity informed generative time series augmentation with few-shot learning for unmanned aerial vehicle battery state-of-health estimation”, *Journal of Energy Storage*, 2026. (**IF: 10.7**) DOI:10.1016/j.est.2026.123752.
68. Liang Cao, **Weide Liu***, Jun Cheng, Yan Qin, Bhushan Gopaluni, “IPM-FM: A Foundation Model with Consensus Feature Selection for Industrial Process Monitoring”, *ICIEA*, 2026.

Professional Services & Awards

Editorial Positions

- Associate Editor, *IEEE Journal of Biomedical and Health Informatics*
- Associate Editor, *Signal, Image and Video Processing*
- Guest Editor, *Journal of Visual Communication and Image Representation*

Conference Organization

- Area Chair: BMVC 2026
- Area Chair (Senior Program Committee): AAAI 2027
- Track Chair: IECON 2023
- Special Session Chair: ICIP 2024–2026
- Special Session Chair: ICME 2025–2026

Awards & Honors

- **DAAD AInet Fellowship 2024** (AI for Science)
- **Best Paper Award**, 4th International Workshop on “Deep Learning for Human Activity Recognition,” IJCAI (First Author)
- **Best Paper Award**, ICAIS&ISAS (Corresponding Author)
- **Best Paper Award**, Companion Proceedings of the ACM Web Conference 2026 (Corresponding Author)
- MICCAI 2022 Travel Grant